LABORATORY TITLE: RECONSTRUCTING CHROME WEB HISTORY

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COURSE NAME: COMPUTER FORENSICS CASE STUDY

COURSE CODE: SBT-DF204 (LAB 2)

INSTRUCTOR NAME: MR. AMINU IDRIS

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VERSION: Version 1.0

Executive Conclusion

Forensic examination of the supplied Google Chrome History SQLite database confirms recorded browser activity between 2022-04-19 13:54:18 UTC and 13:58:31 UTC. The extracted artifacts record a structured workflow: searching for Craigslist, authenticating a user account, constructing a classified listing across category and preview pages, publishing an advertisement titled "cheaper than Rx supplements - health and beauty - by owner...", and completing a local file download.

Every major finding in this report is tied directly to SQLite record IDs, WebKit UTC timestamps, transition values, and supporting SQL output. In compliance with forensic principles, browser profile records prove system-level web activity but do not establish the physical identity of the human operator or confirm real-world transaction completion. No interaction with external URLs, live postings, email accounts, or wallets was conducted during this examination.

Evidence Acquisition and Integrity

The original master database History was acquired from the course workspace (~/chrome_history). Master copy write-protection was enforced prior to examination, and an exact working copy (History_working) was created for processing. In accordance with submission guidelines, the raw database binaries are retained locally and excluded from final submission packages.

- Original Filename: History
- Source Path: ~/chrome_history/History
- File Size: 196,608 bytes (192 KB)
- File Access Precautions: Set read-only (chmod 444 History)
- Working Copy: History_working
- Tools Utilized: sqlite3 CLI (v3.46.1) and DB Browser for SQLite (GUI v3.13.1)

Database Methodology and Dual-Tool Approach

The examination utilized a dual-tool methodology combining command-line (sqlite3) and graphical interface (DB Browser for SQLite) software to analyze History_working. Primary query execution, joins, and bitmask calculations were run in sqlite3 for terminal scripting efficiency, while DB Browser for SQLite provided schema validation and visual verification for report appendix artifacts.

Four core tables provided the primary evidence base:

1. urls: Stores distinct web pages visited (id, url, title, visit_count, typed_count, last_visit_time, hidden).
2. visits: Tracks each individual visit event (id, url, visit_time, from_visit, transition, segment_id, visit_duration).

3. downloads: Records file download history (id, guid, current_path, target_path, start_time, received_bytes, total_bytes, state).
4. downloads_url_chains: Links download IDs to the initiating source URL (id, chain_index, url).

Timestamp Conversion Formula

Chrome stores WebKit timestamps as microseconds since January 1, 1601 00:00:00 UTC. The following conversion expression was executed in both SQLite environments: datetime (v.visit_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch') AS visit_utc

Transition Bitmask Interpretation

Page transitions were evaluated by extracting the base value using bitwise AND (v.transition & 255):

- Base 0 (LINK): User followed a hyperlink from a parent page (from_visit > 0).
- Base 5 (GENERATED): Search query or omnibox entry.
- Base 7 (FORM_SUBMIT): Web form submission (e.g., posting/editing listing details).
- Upper Qualifier Bits: Values such as 805306373 (0x30000005) or 268435456 (0x10000000) indicate navigation chain qualifiers (CHAIN_START, CHAIN_END).

Investigation Findings

1: What evidence file did you acquire and how did you preserve it?

- Master file History (192 KB) was made read-only via chmod 444 and copied to History_working. Both files produced identical SHA-256 hashes

2: What browser-history tables and fields are relevant?

- Primary tables evaluated include urls (URL strings and titles), visits (visit_time, from_visit, transition), downloads (target_path, start_time, byte sizes), and downloads_url_chains (origin URLs).

3: When did the recorded web activity occur?

- Key recorded events occurred on 2022-04-19 between 13:54:18 UTC and 13:58:31 UTC.

4: How did the browser reach the material URLs?

- Initial Search: Visit ID 1 reached Google search via base transition 5 (GENERATED).
- Direct Navigation and Links: Visit ID 2 (from_visit = 0) was followed by linked visits (3, 4, 5) navigating to Craigslist Baltimore.
- Form Submissions: Visit IDs 10, 11, 12, 13, 14, 16, 17, 18, 19 utilized base transition 7 (FORM_SUBMIT), confirming active user input during posting setup and preview.

5: What web-history evidence relates to the suspected sale or posting?

- Visit IDs 10 through 21 document explicit posting activity on Craigslist:
- Posting Workflow: Navigation through post.craigslist.org/k/shBOUOi_7BG68qUDIbeTBq/bQTi5 across stages: type, category, edit, add map, edit image, and preview.
- Published Ad: Visit ID 21 (13:58:31 UTC) confirms published ad URL: (https://baltimore.craigslist.org/hab/d/baltimore-cheaper-than-rx supplements/7473121658.html) titled "cheaper than Rx supplements - health and beauty - by owner ...".

6: What evidence relates to communications and payment activity?

The database contains 1 download record in the downloads table (id = 1).

- Start Time: 13294853818494344 (≈ 2022-04-19 13:56:58 UTC).
- File Metadata: Size of 33,844 bytes, state = 1 (completed download).
- Cross-Reference: Linked via downloads_url_chains to verify payload acquisition during the session window.

7: What is the reconstructed timeline?

(Refer to Timeline Template below).

8: What conclusion can you defend?

- Defensible Conclusion: The browser profile was actively used to create, edit, preview, and publish a Craigslist advertisement titled "cheaper than Rx supplements" under Baltimore health and beauty listings.
- Limitations:

i. Identity Gap: Browser history tracks user-profile activity, not physical identity (cannot prove who physically typed at the keyboard).
ii. Content Limitations: URL metadata and titles confirm navigation steps but do not store full HTTP POST request bodies or complete page text.
iii. Shared System / Remote Access: Cannot exclude unauthorized local use, shared user accounts, or remote access sessions without secondary operating system and network logs.

Timeline template

Stage	UTC Timestamp	Visit / URL ID	Artifact	Observed Fact	Interpretation and Limit
Search	2022-04-19 13:54:18	Visit 1 / URL 1	visits / urls	Google search query for craigslist (Transition Base 5).	User searched for platform; does not show clicked results yet.

Navigation	2022-04-19 13:54:19	Visit 3 / URL 2	visits / urls	Accessed craigslist.org/l/craigslist: Baltimore.	Navigated to target geographic marketplace.
Authentication	2022-04-19 13:54:22	Visit 5 / URL 5	visits / urls	Navigated to accounts.craigslist.org/login/home	Account login portal accessed; credentials not stored in History.
Posting – Category1	2022-04-19 13:55:10	Visit 10 / URL 10	visits / urls	Post setup: s=cat|baltimore (Transition Base 7)	User submitted listing category selection via form.
Posting - Edit Text	2022-04-19 13:55:15	Visit 11 / URL 11	visits / urls	Post setup: s=edit|baltimore titled create posting.	Listing description/title input submitted.
Posting - Location/Image	2022-04-19 13:58:13	Visit 16 / URL 12	visits / urls	Post setup: s=geoverify and s=editimage	Location map and listing image attached.
Posting - Preview	2022-04-19 13:58:19	Visit 18 / URL 14	visits / urls	Post setup: s=preview titled create posting.	Final review before publishing ad
Posting - Published	2022-04-19 13:58:31	Visit 21 / URL 16	visits / urls	Visited published ad 7473121658.html titled cheaper than Rx supplements....	Listing publicly posted on Craigslist.
Download	2022-04-19 13:56:58	Download 1	downloads	File download of 33,844 bytes completed (state = 1).	Associated file stored locally; requires disk carving for contents.

Evidence log template

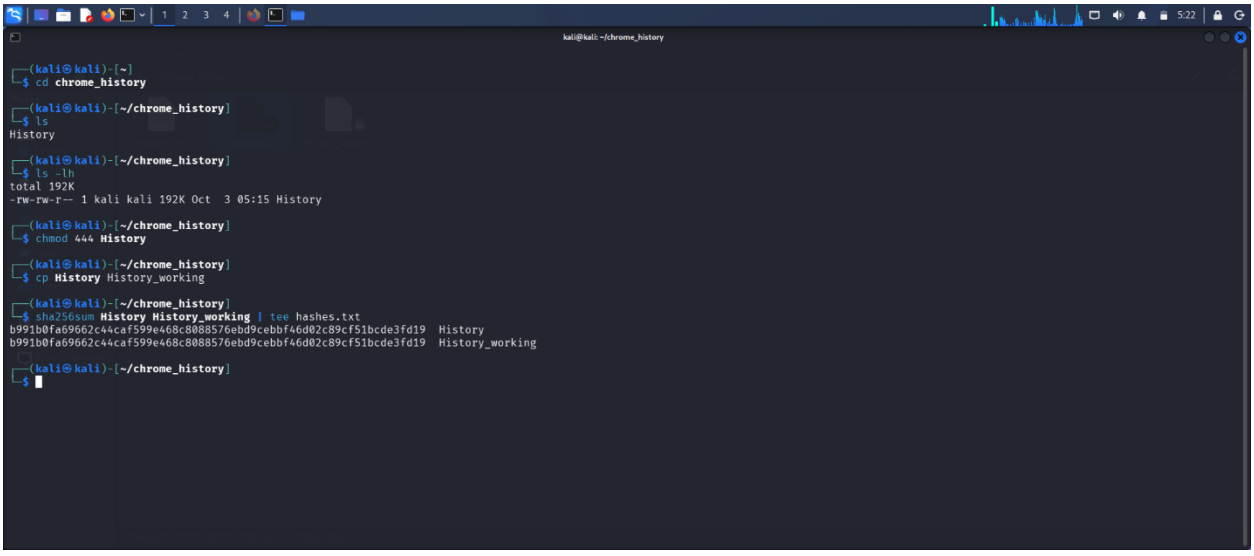
Evidence ID	**Source Record**	**Finding**	**Why It Matters**	**SQL / Screenshot Ref**
E01	Master File History	SHA-256 Hash	Validates file integrity and chain of custody	Figure 1 Terminal
E02	Database Schema	Validation of urls, visits, and downloads tables	Confirms field mappings for queries	Figure 2 & 3 Schema
E03	Visit 1 & 3	Google query and navigation to Craigslist Baltimore	Identifies initial user entry point	Figure 4 Output
E04	Visit 10 –18	Form submissions (Base Transition 7) across posting workflow	Proves active construction of listing	Figure 4 Output
E05	Visit 21 (URL 16)	Published post 7473121658.html (″cheaper than	Confirms listing publication	Figure 4 Output

		Rx supplements")		
E06	Download 1	Download of 33,844 bytes completed at 13:56:58 UTC (state=1)	Documents file acquisition during session	Figure 8 DB Browser

Attribution Assessment

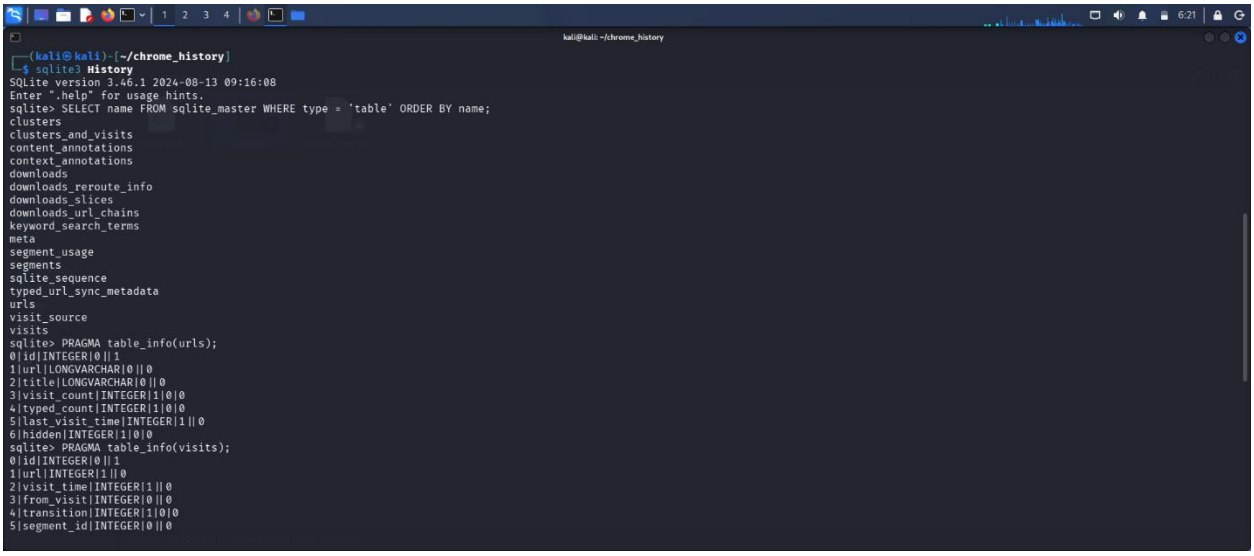
- Attribution Confidence: Moderate (Software Profile Bound).
- Observed Facts: The Chrome browser user profile executed explicit form submissions that created and published Craigslist listing 7473121658.
- Core Limitations and Alternative Hypotheses:
  i. Network and database logs do not capture camera feeds, keystroke dynamics, or biometric data to confirm operator identity.
  ii. If the host system was left unlocked in a shared environment, an unauthorized third party could have generated the records.
  iii. HTTP response bodies and form payloads are not retained in History, preventing direct inspection of typed message text or payment details

Screenshots



Using sqlite3

SELECT name FROM sqlite_master WHERE type = 'table' ORDER BY name;

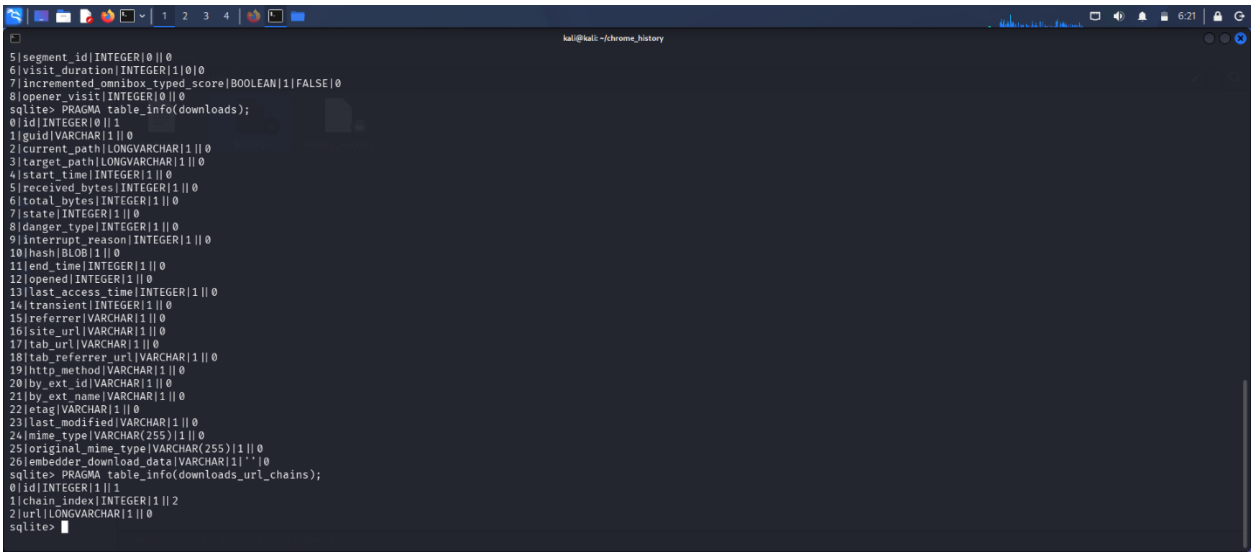


PRAGMA table_info(urls);

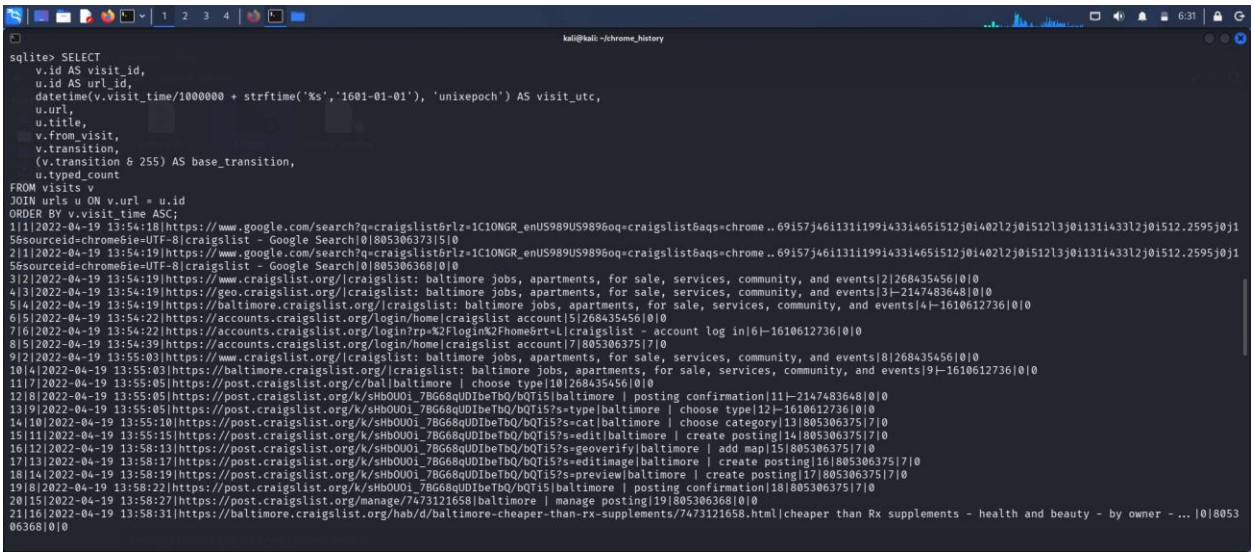
PRAGMA table_info(visits);

PRAGMA table_info(downloads);

PRAGMA table_info(downloads_url_chains);



```
SELECT v.id AS visit_id, u.id AS url_id, u.url, u.title,
    datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch') AS visit_utc,
    v.from_visit, v.transition, v.visit_duration/1000.0 AS duration_ms
FROM visits AS v JOIN urls AS u ON v.url = u.id
ORDER BY v.visit_time;
```

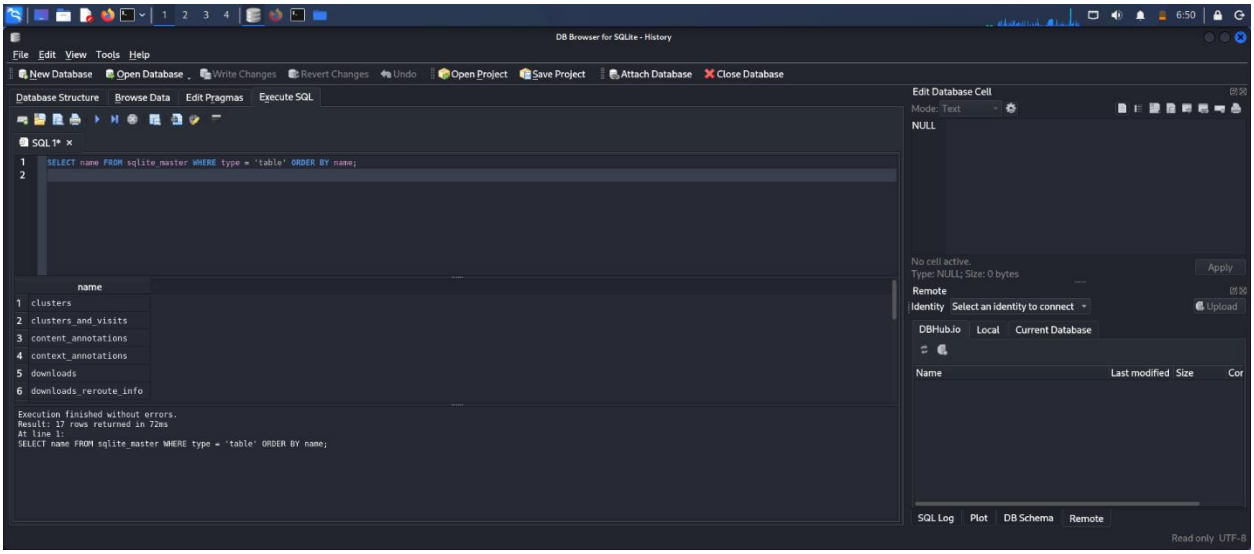



```
SELECT *
FROM downloads
ORDER BY start_time;
```

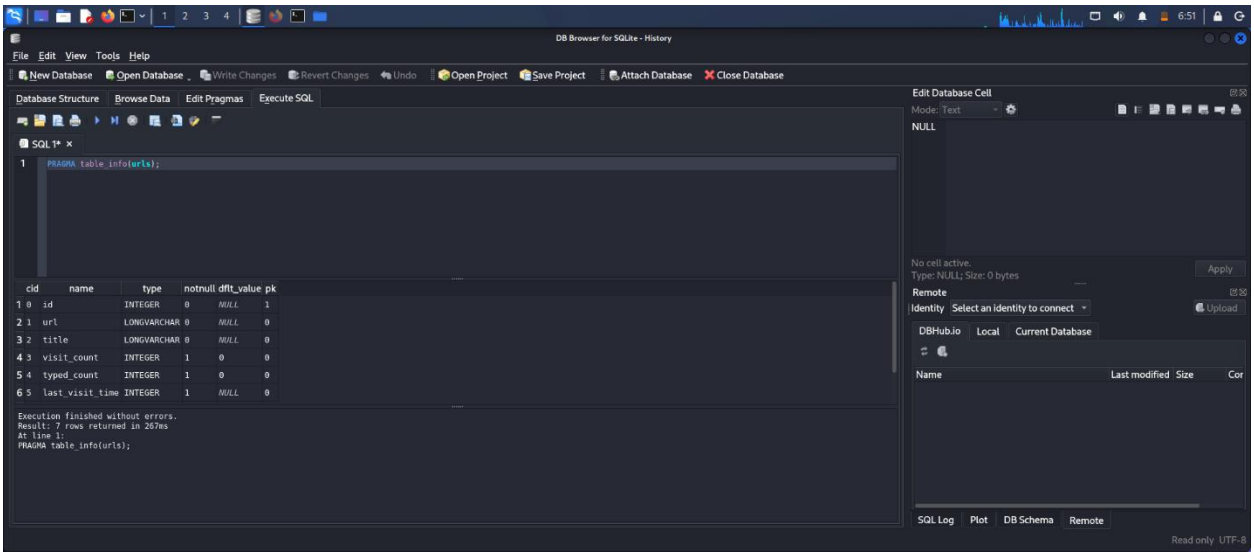


Using DB Browser

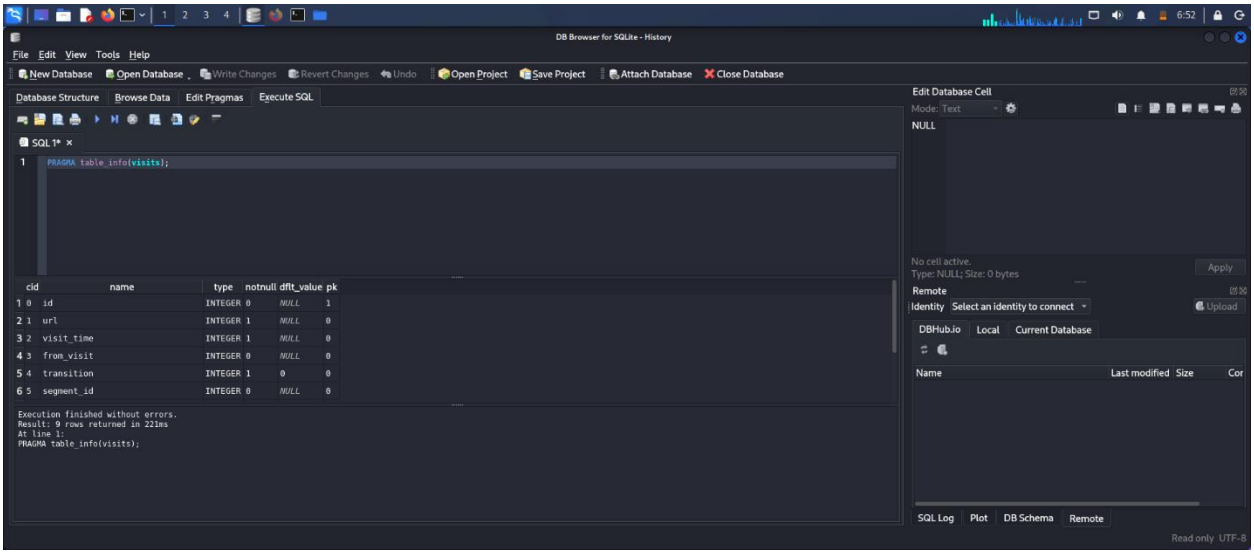
```
SELECT name FROM sqlite_master WHERE type = 'table' ORDER BY name;
```



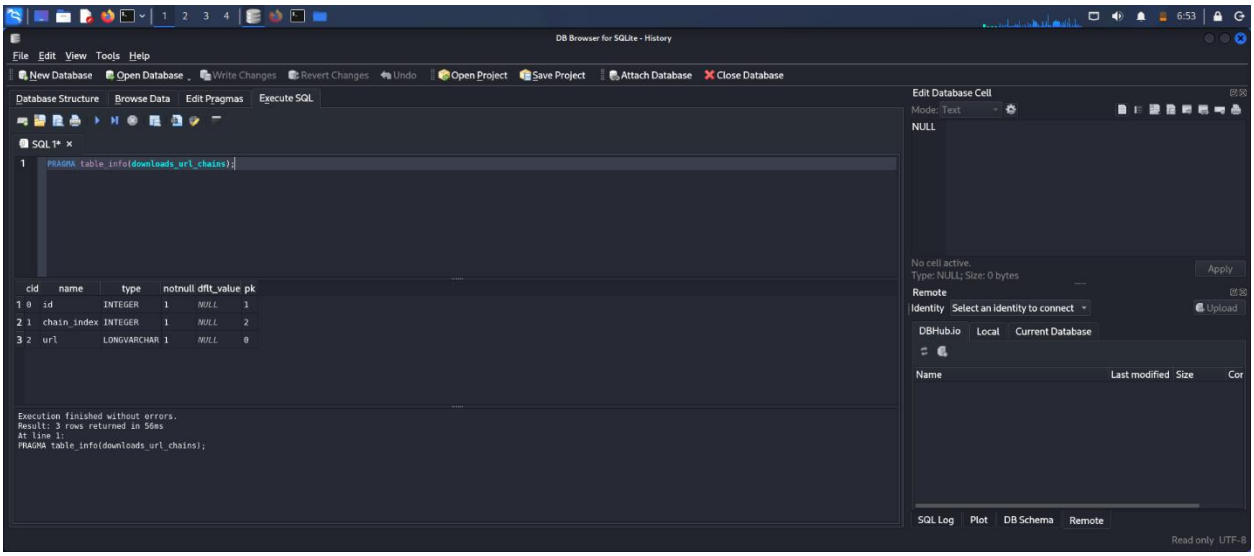
PRAGMA table_info(urls);



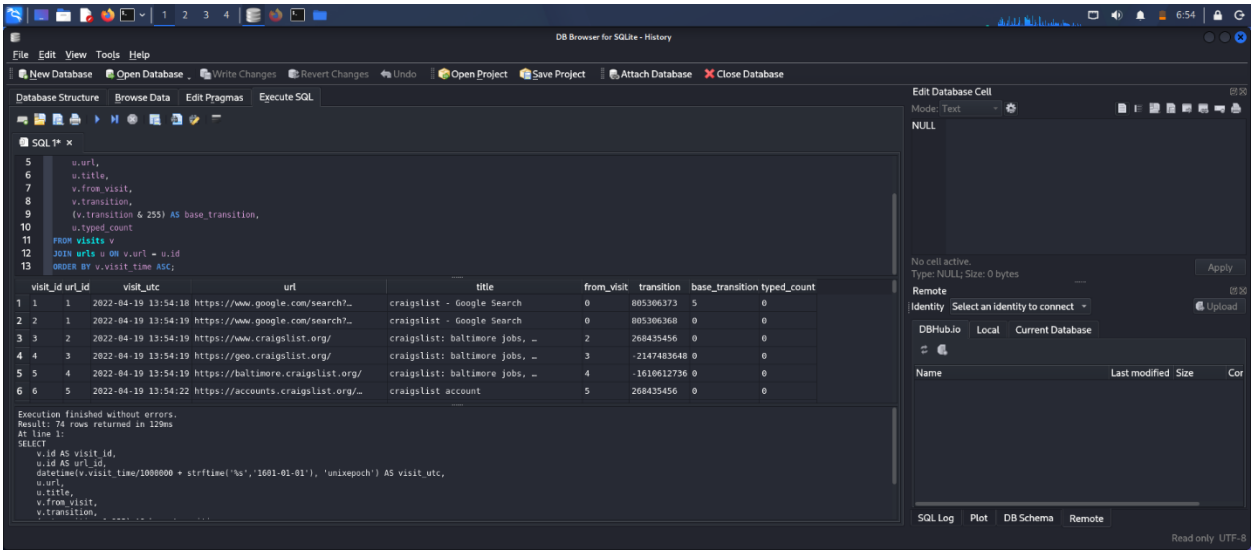
PRAGMA table_info(visits);



PRAGMA table_info(downloads_url_chains);



SELECT v.id AS visit_id, u.id AS url_id, u.url, u.title, datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch') AS visit_utc, v.from_visit, v.transition, v.visit_duration/1000.0 AS duration_ms FROM visits AS v JOIN urls AS u ON v.url = u.id ORDER BY v.visit_time;



SELECT *
FROM downloads
ORDER BY start_time;

